Data-Driven Investment Intelligence Using NIFTY-50 Market Data

Participation Format

This is a team problem statement.

Teams may consist of up to 02 participants.

Participants are expected to collaboratively design and develop an AI-powered investment intelligence platform using the provided NIFTY-50 dataset. All submitted work must be original and completed within the competition timeline.

The challenge evaluates not only machine learning and analytical capabilities but also the ability to transform financial data into practical decision-support systems that can assist investors in making informed decisions.

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1. Motivation

Financial markets generate massive amounts of data every day, making it increasingly difficult for investors to identify meaningful patterns and make informed decisions.

While stock market data is publicly available, transforming raw historical prices and trading volumes into actionable insights requires sophisticated analytical techniques.

The NIFTY-50 index represents fifty of the largest and most influential companies listed on the National Stock Exchange of India, spanning multiple sectors such as Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

This challenge invites participants to build an AI-powered investment intelligence platform capable of extracting insights, identifying opportunities, assessing risks, and supporting data-driven investment decisions using historical market data.

Unlike traditional stock-price prediction competitions, the focus of this challenge is on creating practical decision-support systems that combine machine learning, financial analytics, explainability, and user-centric design.

Dataset Overview

Dataset Link:
https://www.kaggle.com/datasets/rohanrao/nifty50-stock-market-data/data

Additional datasets :

https://www.kaggle.com/datasets/stoicstatic/india-stock-data-nse-1990-2020

(Students are requested to use only the provided datasets)

Participants will work with the **NIFTY-50 Stock Market Dataset**, which contains historical market data for companies that have been part of the NIFTY-50 index of the National Stock Exchange (NSE) of India.

The dataset spans from **January 2000 to April 2021** and covers companies across multiple sectors including Banking, Information Technology, Energy, Consumer Goods, Pharmaceuticals, Financial Services, and Manufacturing.

The dataset consists of:

Historical Stock Data

Daily records for each company containing:

- Open Price
- High Price
- Low Price
- Close Price
- Traded Volume
- Turnover

Company Metadata

Additional information including:

- Company Name
- Sector

- Industry Classification

The dataset provides more than two decades of historical market activity, enabling participants to analyze long-term trends, market cycles, sectoral performance, volatility patterns, and investment opportunities.

Examples of derived indicators include:

- Moving Averages (MA)
- Exponential Moving Averages (EMA)
- Relative Strength Index (RSI)
- MACD
- Bollinger Bands
- Volatility Indicators
- Momentum Indicators

The challenge is designed to evaluate a participant's ability to transform raw historical market data into actionable investment intelligence and decision-support tools.

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2. Problem Statement

You are part of the Data Science and Investment Analytics team of a financial technology company.

Your objective is to develop an intelligent investment intelligence platform capable of transforming historical market data into meaningful insights that can assist investors in making informed decisions.

The platform should help users:

- Analyze historical stock performance.
- Evaluate investment opportunities.
- Assess portfolio risk.
- Understand market trends and behavior.
- Generate actionable investment insights.
- Support data-driven decision making.

Participants are free to use any combination of:

- Machine Learning
- Time Series Forecasting
- Statistical Modeling

- Deep Learning
- Portfolio Optimization
- Financial Analytics

The final solution should go beyond simple stock-price forecasting and focus on delivering practical investment intelligence.

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3. Mandatory and Optional Tasks

Mandatory Tasks

A. Stock Predictor Engine

Develop predictive models capable of forecasting future stock behavior using historical market data.

Examples:

- Predict future stock prices or returns over a specified investment horizon.
- Predict the direction of stock price movement (upward or downward).
- Forecasts should be supported by appropriate model evaluation and validation techniques.

Evaluation Metrics:
Participants may use metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, and Directional Accuracy to assess model performance. Additional evaluation metrics may be incorporated if appropriately justified.

B. Portfolio Construction Module

Generate investment portfolios for different investor profiles.

Examples:

- Conservative Investor
- Balanced Investor
- Aggressive Investor

The system should recommend portfolio allocations and justify its decisions.

C. Risk Assessment Module

Evaluate the historical risk associated with individual stocks or portfolios.

Possible Metrics Include:

- Volatility
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Risk-Adjusted Return

Participants may explore additional risk metrics where appropriate.

Optional Tasks

A. Explainable AI Framework

Provide explanations supporting recommendations and insights generated by the system.

B. Personalized Investment Strategies

Design investment recommendations tailored to different risk tolerances and investment goals.

C. Market Anomaly Detection

Identify unusual patterns or events within historical market data.

Examples:

- Sudden Volatility Spikes
- Extreme Drawdowns
- Unusual Trading Activity

D. Forecasting Module

Develop predictive models for:

- Future Price Trends
- Volatility Estimation
- Return Forecasting

Participants should clearly discuss assumptions and limitations.

E. Deployment

Deploy the solution as:

- A Web Application
- An Interactive Dashboard
- A Cloud-Based Service
- A Local Desktop Application

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4. Constraints

Participants may use only the datasets provided by the organizers.

Allowed

- Feature Engineering
- Statistical Transformations
- Technical Indicators
- Machine Learning Models
- Deep Learning Models
- Portfolio Optimization Techniques

Examples include:

- Moving Averages
- Exponential Moving Averages
- RSI
- MACD
- Bollinger Bands
- Momentum Indicators
- Volatility Indicators

Not Allowed

- Live Market Data
- Financial APIs
- News Datasets
- Social Media Sentiment Data

- Proprietary Financial Data
- Alternative Market Datasets

The objective is to evaluate how effectively participants can leverage the provided dataset.

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5. Deliverables

Deliverable 1: Working Prototype

A functional application demonstrating the implemented capabilities.

Deliverable 2: Technical Report

Format: PDF

The report should include:

- Exploratory Data Analysis (EDA)
- Feature Engineering
- Methodology
- Model Architecture
- Portfolio Construction Logic
- Risk Assessment Methodology
- Explainability Techniques
- Key Insights

Maximum Length: 12 Pages

Deliverable 3: Public GitHub Repository

The repository should contain:

- Source Code
- Model Files
- Configuration Files
- Documentation

The solution should be reproducible.

Deliverable 4: README.md

Include instructions for:

- Environment Setup
- Dependency Installation
- Running the Application
- Reproducing Results

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6. Guidelines

1. Focus on Decision Support

The objective is not simply to predict stock prices but to assist users in making informed investment decisions.

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2. Justify Your Recommendations

All investment recommendations and portfolio suggestions should be supported by quantitative evidence.

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3. Emphasize Explainability

Users should be able to understand why a recommendation or insight was generated.

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4. Balance Complexity and Usability

Sophisticated analytics should be presented in a manner that remains understandable and actionable.

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5. Encourage Innovation

Participants are encouraged to explore novel approaches for investment intelligence, risk assessment, portfolio optimization, and explainable AI.

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7. Evaluation Criteria

Criterion	Weightage
Quality of Investment Insights	25%
Technical Innovation	20%
Portfolio & Risk Analytics	20%
Explainability & Transparency	20%
Reproducibility & Documentation	15%
